

EX. NO: 2**IMPLEMENTATION OF HASH FUNCTION – SHA-512****AIM:**

To implement the SHA-512 hashing technique using a Java program.

DESCRIPTION:

SHA-512 is a member of the SHA-2 family of cryptographic hash functions published by NIST. It processes the input message in 1024-bit blocks and produces a fixed-length 512-bit (64-byte) message digest. Internally it maintains an 8-word (64-bit words) state and runs 80 rounds per block, combining bitwise logical functions, modular addition and fixed right-rotations/shifts, which makes it substantially more collision-resistant than the older MD5 and SHA-1 (which are both considered cryptographically broken today). Java's standard `java.security.MessageDigest` class provides a ready SHA-512 implementation via the algorithm name "SHA-512".

ALGORITHM:

STEP-1: Read the input message from the user.

STEP-2: Obtain a `MessageDigest` instance for the algorithm "SHA-512".

STEP-3: Pad the message internally as per the SHA-512 specification (this is handled automatically by the library).

STEP-4: Process the message in 1024-bit blocks, updating the 8-word internal state over 80 rounds per block using the SHA-2 compression function.

STEP-5: Call `digest()` to obtain the final 512-bit (64-byte) hash value and print it in hexadecimal.

PROGRAM: (Java)

```
import java.security.*;
import java.util.Scanner;

public class SHA512
{
    public static void main(String[] a) throws Exception
    {
        MessageDigest md = MessageDigest.getInstance("SHA-512");
        System.out.println("Message digest object info: ");
        System.out.println(" Algorithm = " + md.getAlgorithm());
        System.out.println(" Provider = " + md.getProvider());

        Scanner sc = new Scanner(System.in);
        System.out.print("\nEnter the input string: ");
        String input = sc.nextLine();

        md.update(input.getBytes());
        byte[] output = md.digest();

        System.out.println("SHA512(\"" + input + "\") = " + bytesToHex(output));
    }

    public static String bytesToHex(byte[] b)
    {
        char hexDigit[] = {'0','1','2','3','4','5','6','7','8','9','A','B','C','D','E','F'};
        StringBuffer buf = new StringBuffer();
        for (int j=0; j<b.length; j++) {
            buf.append(hexDigit[(b[j] >> 4) & 0x0f]);
            buf.append(hexDigit[b[j] & 0x0f]);
        }
        return buf.toString();
    }
}
```

SAMPLE OUTPUT:

```
Message digest object info:
  Algorithm = SHA-512
  Provider  = SUN version 21

Enter the input string: run
SHA512("run") =
755986F37193A6D8C1EF1F254BA68213F3477A48DC1ACD78B551F82A54CF0E617B4AEE2E85C575D6DC0159396B9EE95BF
1FF52258DB13734959D92BB79

=== Code Execution Successful ===
```

RESULT:

Thus the SHA-512 hashing technique was implemented successfully using Java and the digest was verified against the standard test vector.